

DUSSEHRA HOLIDAY AI PROJECT CHALLENGE - CLASS 10

AI Creator Challenge | 10-Day Practical & Creative Holiday Work

Learn • Explore • Create • Test • Improve

Choose an ONE project. These projects are limited to Units 1, 2, 3, 4 and 7

1. Data Detective

Idea: Use a small real-world dataset and investigate what the data reveals.

What to do:

- Collect or use a simple dataset such as a school canteen survey.
- Organize and clean the data.
- Use Python/Jupyter, Pandas, and Matplotlib if possible.
- Create graphs and explain the patterns you found.

Final output: A mini data analysis project with dataset, graphs and conclusions.

Syllabus units: Unit 2 + Unit 7

2. Can Probability Predict the Future?

Idea: Turn probability into a Python experiment using coins, dice, or another simple random process.

What to do:

- Run a small experiment and then a larger one.
- Compare expected probability with observed results.
- Repeat the experiment if needed.
- Visualize the results and explain what has changed.

Final output: A Python probability experiment with results, graphs and short conclusions.

Syllabus units: Unit 3 + Unit 7

3. Can My AI Be Trusted?

Idea: Investigate how well an AI/classification system performs instead of simply accepting its predictions.

What to do:

- Run a simple classification experiment.
- Record corrects and incorrect predictions.
- Create or interpret a confusion matrix.
- Explore accuracy, precision, recall and F1 score where appropriate.
- Improve the data/model and compare the results.

Final output: A before-and-after AI evaluation report with results and explanation.

Syllabus units: Unit 1 + Unit 2 + Unit 3

Difficulty: Advanced

4. Generative AI Creator Challenge

Idea: Create an original mini-project using Generative AI and investigate how prompts affect results.

What to do:

- Choose a theme such as a future city, robot teacher, space world, story or poster.
- Create several outputs using different prompts.
- Compare the outputs.
- Discuss one or two limitations or responsible-use concerns.

Final output: A Generative AI showcase containing prompts, outputs and reflections.

Syllabus units: Unit 1 + Unit 4

5. Build Your Own Mini AI Tool in Python

Idea: Build one small, useful Python program connected to AI, data or probability.

What to do:

- Choose one: Study Data Analyzer, Student Marks Analyzer, Probability Simulator, Simple Text Sentiment Tool, or another teacher-approved idea.
- Write and test the program.
- Improve it after testing.
- Be prepared to explain the important parts of your code.

Final output: A working Python mini tool plus source code and a short demonstration.

Syllabus units: Unit 1 + Unit 2 + Unit 3 + Unit 7

 Suggested 10-Day Work Plan

Days	Work
Days 1–2	Choose the project, understand the challenge, and make a simple plan.
Days 3–5	Collect data / create content / build the first version.
Days 6–7	Test the project and record mistakes or interesting results.
Days 8–9	Improve the project and prepare the final presentation/demo.
Day 10	Final check: Can you explain what you made, how it works, and what you learned?

★ What You Should Submit

- Completed project / working prototype / poster / digital output.
- A short explanation of what they did and what they discovered.
- Evidence of testing: examples, screenshots, graphs, results, or observations.
- Students should be able to explain their own work. Copying a ready-made project is discouraged.

Simple Evaluation — 20 Marks

Criteria	Marks
Understanding & explanation	5
Creativity / originality	5
Working / completed project	5
Testing, observations & presentation	5

Goal: Don't just learn AI — build, test and improve it! 